

```
In [2]: import pandas as pd
df=pd.read_csv(r'C:\Users\Sowndarya\Downloads\customer_shopping_behavior.csv')

# by default shows 1st 5 rows
df.head()
```

Out[2]:

	Customer ID	Age	Gender	Item Purchased	Category	Purchase Amount (USD)	Location	Size	Color
0	1	55	Male	Blouse	Clothing	53	Kentucky	L	Green
1	2	19	Male	Sweater	Clothing	64	Maine	L	Marc
2	3	50	Male	Jeans	Clothing	73	Massachusetts	S	Marc
3	4	21	Male	Sandals	Footwear	90	Rhode Island	M	Marc
4	5	45	Male	Blouse	Clothing	49	Oregon	M	Turque

In [8]: df.info()

```
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 3900 entries, 0 to 3899
Data columns (total 18 columns):
#   Column                                Non-Null Count  Dtype
---  -
0   Customer ID                          3900 non-null   int64
1   Age                                  3900 non-null   int64
2   Gender                               3900 non-null   object
3   Item Purchased                       3900 non-null   object
4   Category                             3900 non-null   object
5   Purchase Amount (USD)                3900 non-null   int64
6   Location                             3900 non-null   object
7   Size                                  3900 non-null   object
8   Color                                3900 non-null   object
9   Season                               3900 non-null   object
10  Review Rating                        3863 non-null   float64
11  Subscription Status                  3900 non-null   object
12  Shipping Type                        3900 non-null   object
13  Discount Applied                     3900 non-null   object
14  Promo Code Used                      3900 non-null   object
15  Previous Purchases                   3900 non-null   int64
16  Payment Method                       3900 non-null   object
17  Frequency of Purchases                3900 non-null   object
dtypes: float64(1), int64(4), object(13)
memory usage: 548.6+ KB
```

```
In [9]: df.describe()
# summary statistics of numerical columns
```

Out[9]:

	Customer ID	Age	Purchase Amount (USD)	Review Rating	Previous Purchases
count	3900.000000	3900.000000	3900.000000	3863.000000	3900.000000
mean	1950.500000	44.068462	59.764359	3.750065	25.351538
std	1125.977353	15.207589	23.685392	0.716983	14.447125
min	1.000000	18.000000	20.000000	2.500000	1.000000
25%	975.750000	31.000000	39.000000	3.100000	13.000000
50%	1950.500000	44.000000	60.000000	3.800000	25.000000
75%	2925.250000	57.000000	81.000000	4.400000	38.000000
max	3900.000000	70.000000	100.000000	5.000000	50.000000

In [10]: `df.describe(include='all')` *# for character statistics also*

Out[10]:

	Customer ID	Age	Gender	Item Purchased	Category	Purchase Amount (USD)	Location
count	3900.000000	3900.000000	3900	3900	3900	3900.000000	3900
unique	NaN	NaN	2	25	4	NaN	50
top	NaN	NaN	Male	Blouse	Clothing	NaN	Montana
freq	NaN	NaN	2652	171	1737	NaN	96
mean	1950.500000	44.068462	NaN	NaN	NaN	59.764359	NaN
std	1125.977353	15.207589	NaN	NaN	NaN	23.685392	NaN
min	1.000000	18.000000	NaN	NaN	NaN	20.000000	NaN
25%	975.750000	31.000000	NaN	NaN	NaN	39.000000	NaN
50%	1950.500000	44.000000	NaN	NaN	NaN	60.000000	NaN
75%	2925.250000	57.000000	NaN	NaN	NaN	81.000000	NaN
max	3900.000000	70.000000	NaN	NaN	NaN	100.000000	NaN

In [4]: `df.isnull().sum()` *# gives sum of null values*

```
Out[4]: Customer ID      0
        Age             0
        Gender          0
        Item Purchased  0
        Category        0
        Purchase Amount (USD) 0
        Location        0
        Size            0
        Color           0
        Season          0
        Review Rating   37
        Subscription Status 0
        Shipping Type   0
        Discount Applied 0
        Promo Code Used 0
        Previous Purchases 0
        Payment Method  0
        Frequency of Purchases 0
        dtype: int64
```

```
In [5]: df['Review Rating']=df.groupby('Category')['Review Rating'].transform(lambda x:
```

```
In [6]: df.isnull().sum()
```

```
Out[6]: Customer ID      0
        Age             0
        Gender          0
        Item Purchased  0
        Category        0
        Purchase Amount (USD) 0
        Location        0
        Size            0
        Color           0
        Season          0
        Review Rating   0
        Subscription Status 0
        Shipping Type   0
        Discount Applied 0
        Promo Code Used 0
        Previous Purchases 0
        Payment Method  0
        Frequency of Purchases 0
        dtype: int64
```

```
In [7]: df.columns=df.columns.str.lower() # converting column names into lowercase stri
        df.columns=df.columns.str.replace(' ','_') # replace space with '_'
        df.columns
```

```
Out[7]: Index(['customer_id', 'age', 'gender', 'item_purchased', 'category',
              'purchase_amount_(usd)', 'location', 'size', 'color', 'season',
              'review_rating', 'subscription_status', 'shipping_type',
              'discount_applied', 'promo_code_used', 'previous_purchases',
              'payment_method', 'frequency_of_purchases'],
              dtype='object')
```

```
In [21]: df=df.rename(columns={'purchase_amount_(usd)': 'purchase_amount'})
```

```
In [22]: df.columns
```

```
Out[22]: Index(['customer_id', 'age', 'gender', 'item_purchased', 'category',
              'purchase_amount', 'location', 'size', 'color', 'season',
              'review_rating', 'subscription_status', 'shipping_type',
              'discount_applied', 'promo_code_used', 'previous_purchases',
              'payment_method', 'frequency_of_purchases'],
              dtype='object')
```

```
In [23]: # creating age group using qcut in pandas
label=['young adult','adult','middle aged','senior']
df['age_group']=pd.qcut(df['age'],q=4,labels=label) # qcut will cut age into 4
```

```
In [24]: df[['age','age_group']].head(10) #head(10) means first 10 rows
```

```
Out[24]:
```

	age	age_group
0	55	middle aged
1	19	young adult
2	50	middle aged
3	21	young adult
4	45	middle aged
5	46	middle aged
6	63	senior
7	27	young adult
8	26	young adult
9	57	middle aged

```
In [25]: frequency_mapping={'Fortnightly':14,
                             'Weekly':7,
                             'Annually':365,
                             'Monthly':30,
                             'Quarterly':90,
                             'Bi-Weekly':14,
                             'Every 3 Months':90
                           }
df['purchase_frequency_days']=df['frequency_of_purchases'].map(frequency_mappin
```

```
In [26]: df[['purchase_frequency_days','frequency_of_purchases']].head(10)
```

```

-----
KeyError                                Traceback (most recent call last)
Cell In[26], line 1
----> 1 df[[ , ]].head(10)

File ~\anaconda3\Lib\site-packages\pandas\core\frame.py:4119, in DataFrame.__getitem__(self, key)
    4117     if is_iterator(key):
    4118         key = list(key)
-> 4119     indexer = self.columns._get_indexer_strict(key, )[1]
    4121 # take() does not accept boolean indexers
    4122 if getattr(indexer, "dtype", None) == bool:

File ~\anaconda3\Lib\site-packages\pandas\core\indexes\base.py:6212, in Index._get_indexer_strict(self, key, axis_name)
    6209 else:
    6210     keyarr, indexer, new_indexer = self._reindex_non_unique(keyarr)
-> 6212 self._raise_if_missing(keyarr, indexer, axis_name)
    6214 keyarr = self.take(indexer)
    6215 if isinstance(key, Index):
    6216     # GH 42790 - Preserve name from an Index

File ~\anaconda3\Lib\site-packages\pandas\core\indexes\base.py:6264, in Index._raise_if_missing(self, key, indexer, axis_name)
    6261     raise KeyError(f"None of [{key}] are in the [{axis_name}]")
    6263 not_found = list(ensure_index(key)[missing_mask.nonzero()[0]].unique())
-> 6264 raise KeyError(f"{not_found} not in index")

KeyError: "['purchase_frequency_days'] not in index"

```

In [27]: `df[['discount_applied', 'promo_code_used']].head(10)`

Out[27]:

	discount_applied	promo_code_used
0	Yes	Yes
1	Yes	Yes
2	Yes	Yes
3	Yes	Yes
4	Yes	Yes
5	Yes	Yes
6	Yes	Yes
7	Yes	Yes
8	Yes	Yes
9	Yes	Yes

In [50]: `# checking whether 'Discount Applied' and 'Promo Code Used' has same value if ye
(df['discount_applied']==df['promo_code_used']).all()`

Out[50]: `np.True_`

In [28]: `df=df.drop('promo_code_used',axis=1)`

```
In [29]: df.columns # 'promo_code_used' is deleted
```

```
Out[29]: Index([      'customer_id',      'age',
                'gender',      'item_purchased',
                'category',    'purchase_amount',
                'location',      'size',
                'color',      'season',
                'review_rating',  'subscription_status',
                'shipping_type',  'discount_applied',
                'previous_purchases',  'payment_method',
                'frequency_of_purchases',  'age_group',
                ('purchase_frequency_days',)],
              dtype='object')
```

```
In [30]: !pip install mysql-connector-python # to connect mysql with jupyter
```

```
ERROR: Invalid requirement: '#': Expected package name at the start of dependency
specifier
#
^
```

```
In [31]: !pip install pymysql sqlalchemy # to connect mysql with jupyter
```

```
ERROR: Invalid requirement: '#': Expected package name at the start of dependency
specifier
#
^
```

```
In [33]: from sqlalchemy import create_engine
from urllib.parse import quote_plus

username = "root"
password = quote_plus("Sound@123")
host = "localhost"
port = "3306"
database = "customer_behavior"

engine = create_engine(
    f"mysql+pymysql://{username}:{password}@{host}:{port}/{database}"
)

with engine.connect() as conn:
    print("Connected successfully!")

df.to_sql("customers", engine, if_exists="replace", index=False)

print("Data successfully loaded!")
```

```
Connected successfully!
Data successfully loaded!
```

```
In [ ]:
```

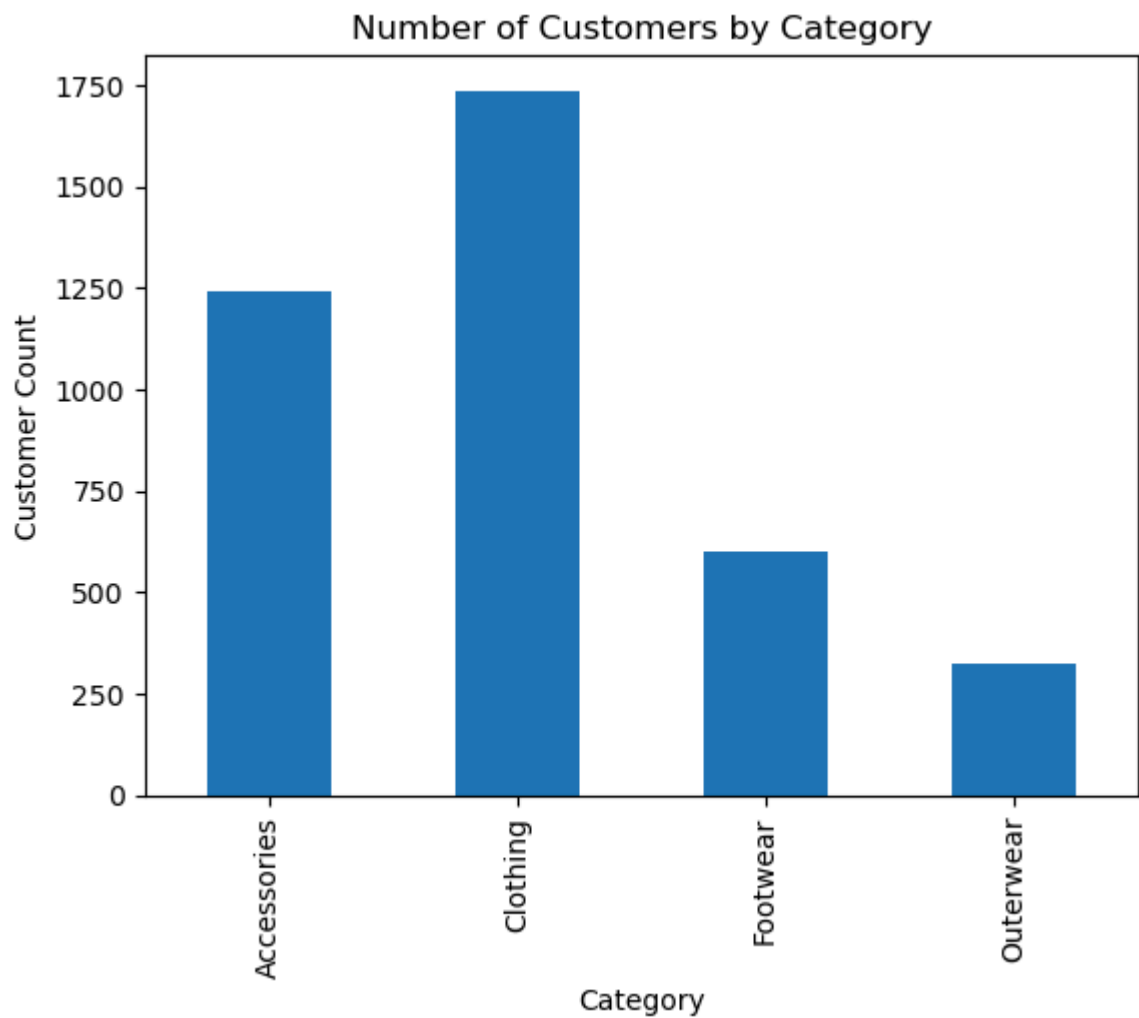
```
In [ ]:
```

```
In [ ]:
```

```
In [9]: import matplotlib.pyplot as plt
```

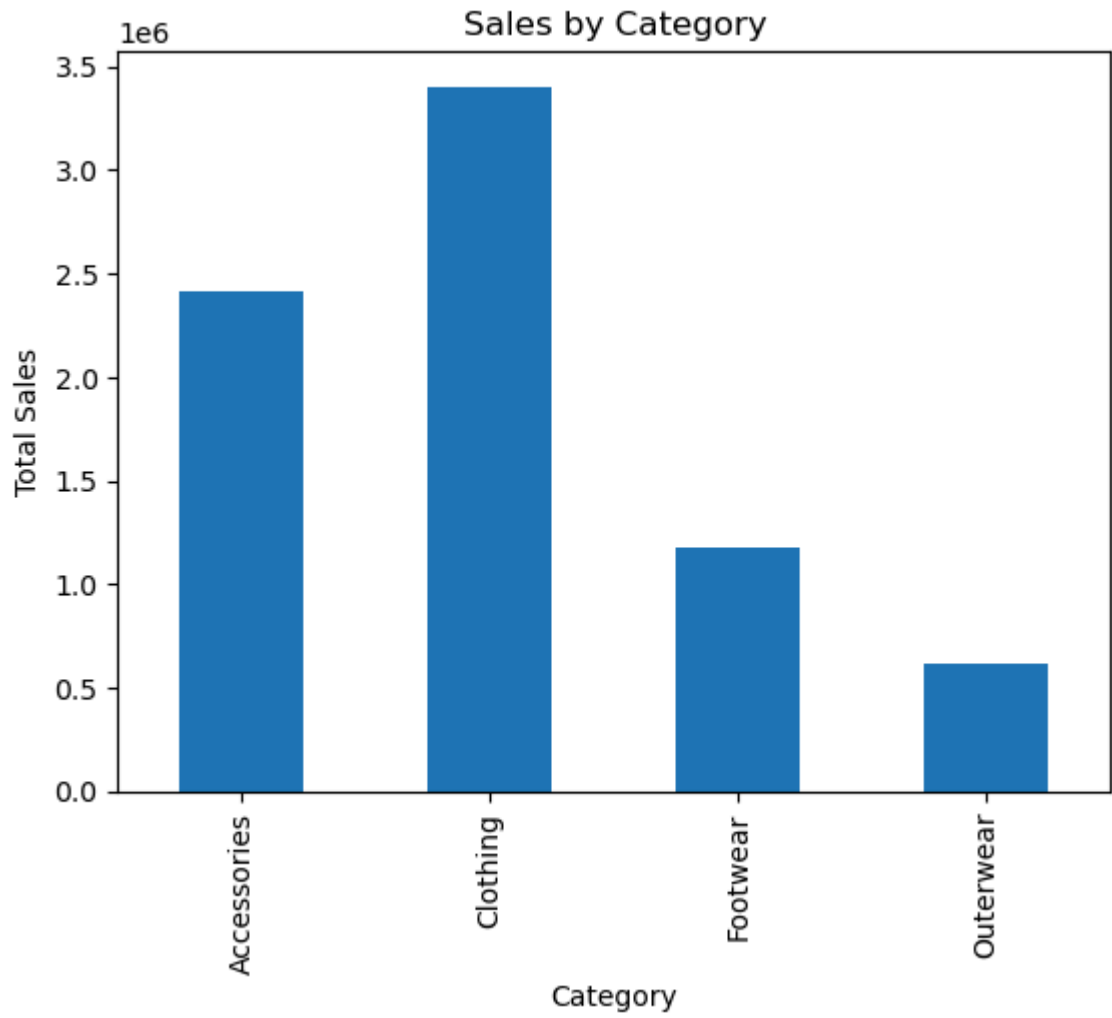
```
df.groupby('category')['customer_id'].count().plot(kind='bar')

plt.title('Number of Customers by Category')
plt.xlabel('Category')
plt.ylabel('Customer Count')
plt.show()
```



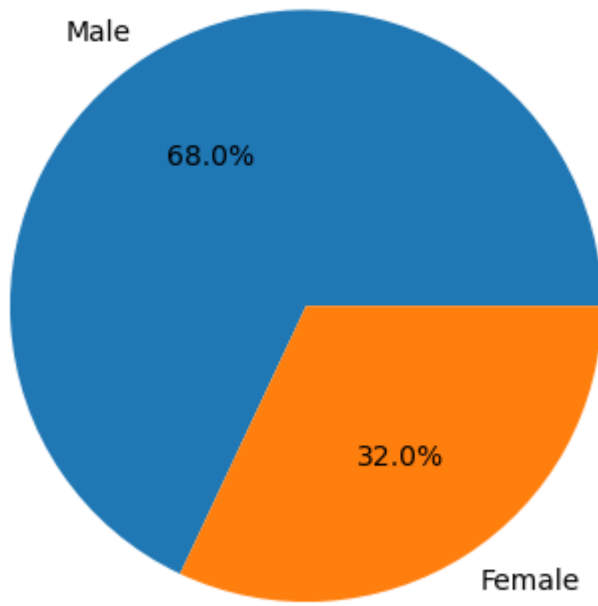
```
In [10]: import matplotlib.pyplot as plt

df.groupby('category')['customer_id'].sum().plot(kind='bar')
plt.title('Sales by Category')
plt.xlabel('Category')
plt.ylabel('Total Sales')
plt.show()
```



```
In [11]: df['gender'].value_counts().plot(kind='pie', autopct='%1.1f%%')
plt.title('Customer Distribution by Gender')
plt.ylabel('')
plt.show()
```


Customer Distribution by Gender



In []: